

# TuniCars+

## Resilience Testing Report

Docker Swarm vs Kubernetes

Session ID

20260813\_095437\_742e19

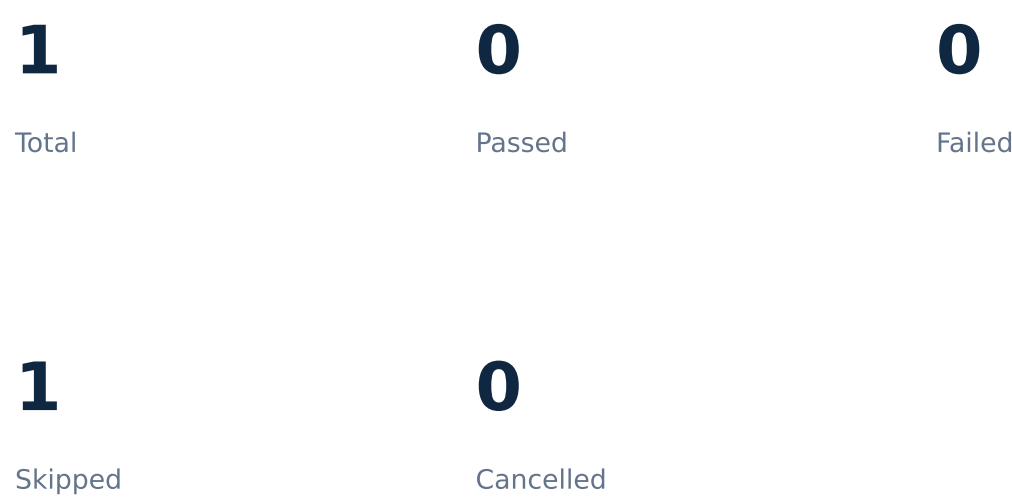
Date

2026-08-13T08:54:37.325210+00:00

# Executive Summary

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Overall availability: N/A

During this campaign, 0 of 1 scenarios passed, 0 failed and 1 were skipped. Conclusions are limited to the measured scenarios and recorded environment constraints.

# Test Environment

Architecture and technologies

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## Frontend → Backend → PostgreSQL

Docker Swarm environment

- Docker services
- FastAPI backend
- Next.js frontend
- PostgreSQL database

Kubernetes / Minikube environment

- Kubernetes Deployments and Services
- Minikube nodes
- FastAPI · Next.js · PostgreSQL

# Test Scenarios

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| Scenario | kubernetes |
|----------|------------|
| latency  | 1          |

# Full Results Table

|          |            |           |          |          |         |              |              |           |              |         | Database probe routes require authentication, but neither RESILIENCE_API_TOKEN nor ESILIENCE_TEST_IDENTIFIER/PASSWORD is configured. |               |
|----------|------------|-----------|----------|----------|---------|--------------|--------------|-----------|--------------|---------|--------------------------------------------------------------------------------------------------------------------------------------|---------------|
| Scenario | Platform   | Detection | Recovery | Required | Success | Avg ms       | P95 ms       | HTTP Fail | Avail. %     | Status  | Skip Reason                                                                                                                          | Error Message |
| latency  | kubernetes | N/A       | N/A      | N/A      | N/A     | Not Measured | Not Measured | N/A       | Not Measured | SKIPPED | Observed: /cars=public:HTTP 401, /renters=public:HTTP 401, /rentals=public:HTTP 401                                                  | N/A           |

# kubernetes — latency

Status: SKIPPED  
Detection Time (s): N/A

Recovery Time (s): N/A

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HTTP Success: N/A

HTTP Failures: N/A

Availability (%): Not Measured

CPU (%): N/A

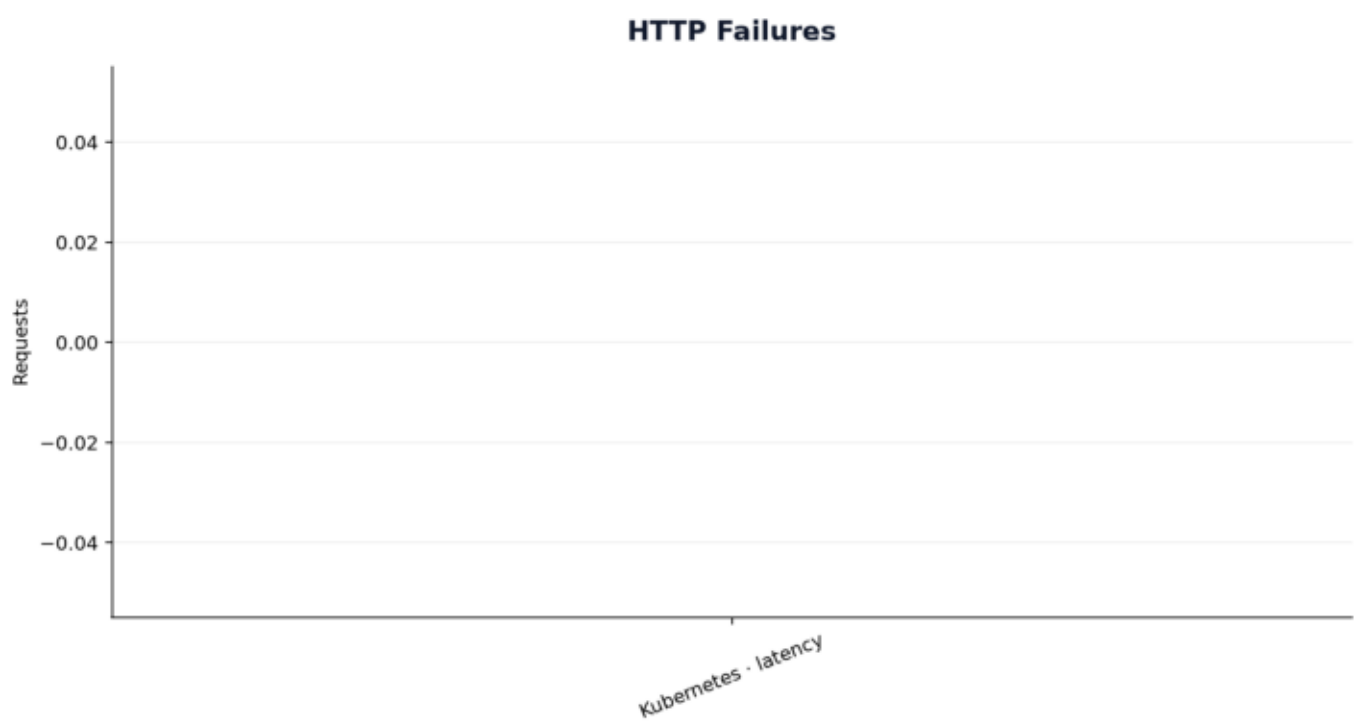
Memory (MB): N/A

## Recorded interpretation

Database probe routes require authentication, but neither RESILIENCE\_API\_TOKEN nor RESILIENCE\_TEST\_IDENTIFIER/RESILIENCE\_TEST\_PASSWORD is configured. Observed: /cars=public:HTTP 401, /renters=public:HTTP 401, /rentals=public:HTTP 401

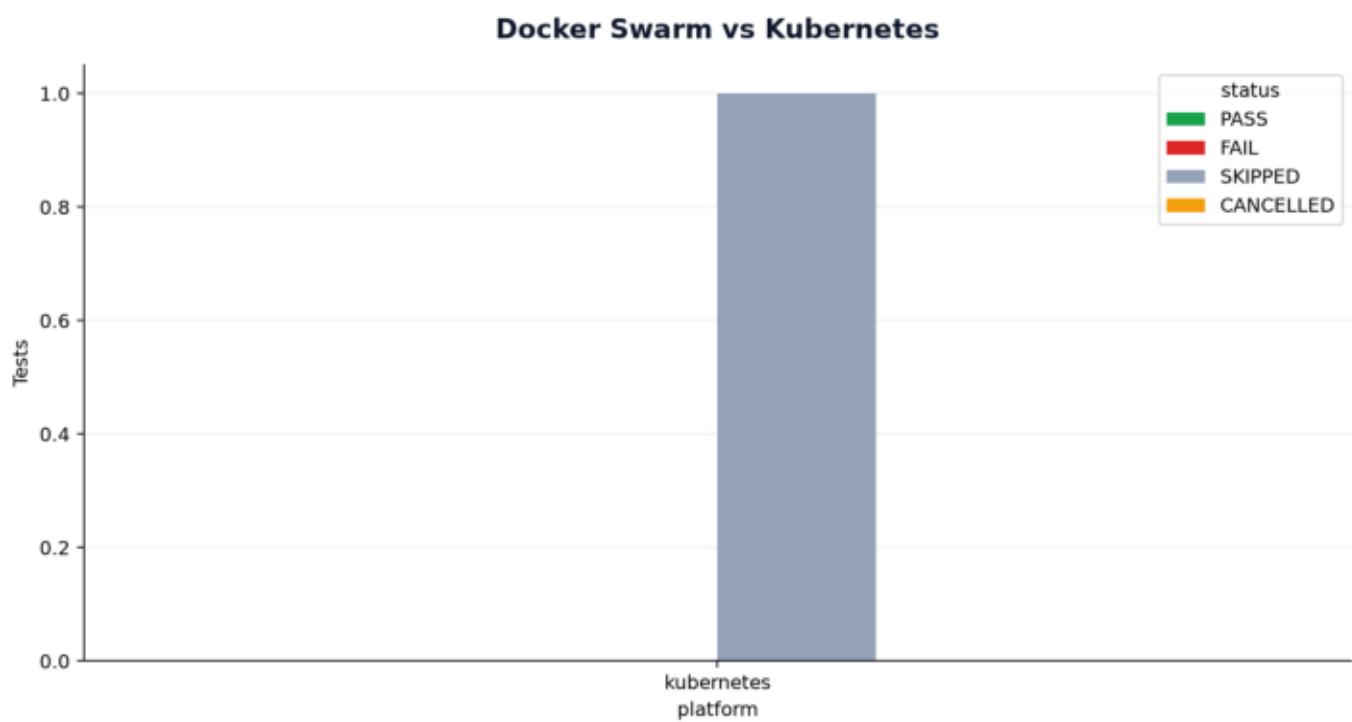
# Http Failures

Current-session measured data



# Platform Comparison

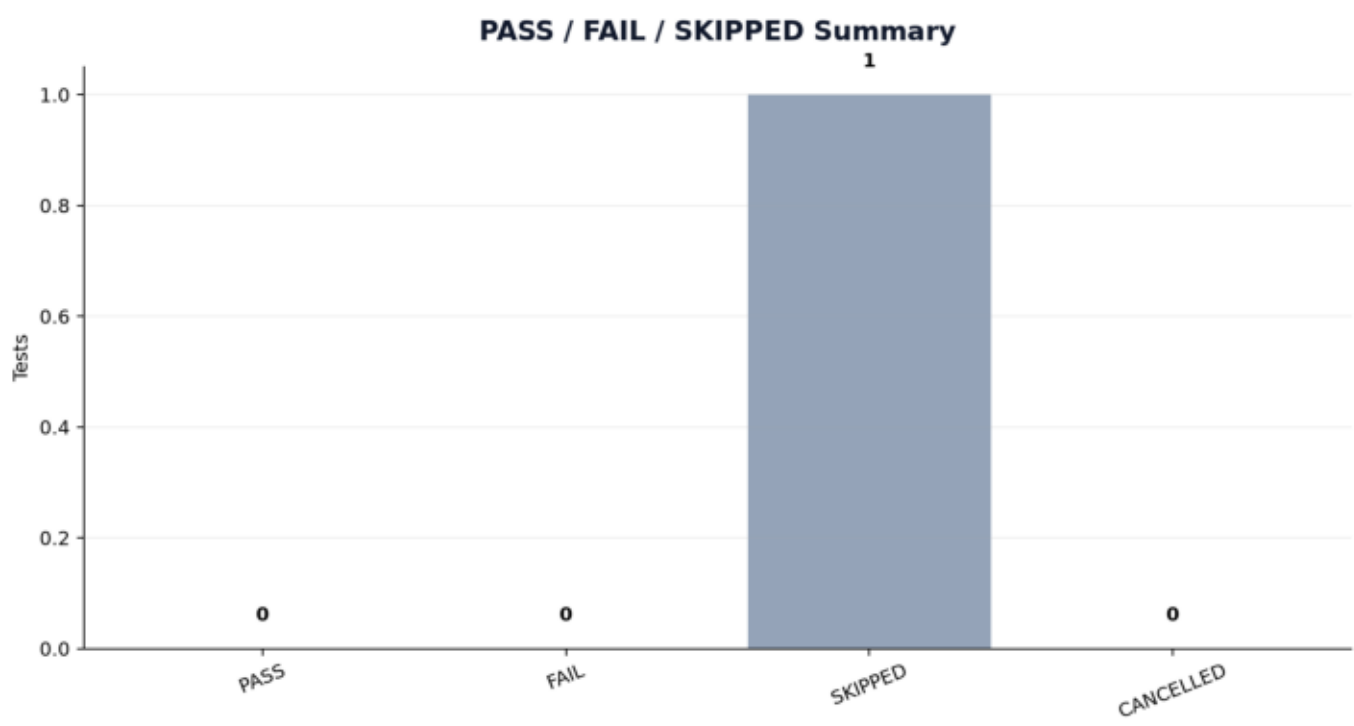
Current-session measured data





# Status Summary

Current-session measured data



# Failures and Errors

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| Platform | Scenario | Status | Error                 |
|----------|----------|--------|-----------------------|
| —        | —        | —      | No failures recorded. |

# Skipped Tests

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| Platform   | Scenario | Exact Reason                                                                                                                                                                                                                                 |
|------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| kubernetes | latency  | Exact Reason: The routes require authentication, but neither RESILIENCE_API_TOKEN nor RESILIENCE_TEST_IDENTIFIER/RESILIENCE_TEST_PASSWORD is configured. Observed: /cars=public:HTTP 401, /renters=public:HTTP 401, /rentals=public:HTTP 401 |

# Swarm vs Kubernetes Comparison

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| Platform   | SKIPPED |
|------------|---------|
| kubernetes | 1       |

# Final Conclusion

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During this campaign, 0 of 1 scenarios passed, 0 failed and 1 were skipped. Conclusions are limited to the measured scenarios and recorded environment constraints.